

Instructor Key: Day 2 Mastery Quiz

Variables, Expressions, and State

Key for the Day 2 student quiz package.

1 Artifact status and source boundary

Field	Status
Student artifact	10-point Day 2 mastery quiz; quiz, not workbooklet.
Content anchor	Built from the Day 2 workbooklet content surface: current value, planned value, assignment history, expression value, overwrite, approval checkpoint, and lab-log correction.
Target	Variables, assignment, arithmetic expressions, current state, planned versus recorded state, state history, and checkpoint-safe explanation.
Scenario/source boundary	Lower-sensitivity pump-calibration record. No protected exam/review prompt text, answer-key language, private student data, or near-copy source structure used.
Student-surface rule	Student PDF intentionally hides internal file versioning and release/validation language; visible student surface carries Lance copyright.
Canvas status	Not Canvas-released. Student PDF still needs Lance upload/staging plus Ally and Panorama review if selected for use.
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2 Reference code

```
1 # Cell A: baseline pump record
2 pump_id = "P-18"
3 target_flow_lpm = 20.0
4 flow_lpm = 18.5
5 gap_lpm = target_flow_lpm - flow_lpm
6
7 # Cell B: calculate first planned correction
8 adjust_lpm = 1.0
9 planned_flow_lpm = flow_lpm + adjust_lpm
10 planned_gap_lpm = target_flow_lpm - planned_flow_lpm
11
12 # Cell C: approve the first planned correction
13 flow_lpm = planned_flow_lpm
14 gap_lpm = planned_gap_lpm
15
16 # Cell D: calculate a fine adjustment, but do not record it yet
17 adjust_lpm = 0.5
18 planned_flow_lpm = flow_lpm + adjust_lpm
19 gap_lpm = target_flow_lpm - planned_flow_lpm
20
21 # Cell E: record the fine adjustment after approval
22 flow_lpm = planned_flow_lpm
23 gap_lpm = target_flow_lpm - flow_lpm
```

3 Point map, secure answers, and support moves

Pt.	Item target	Secure answer	Common issue / support move	Tags
1	Current-flow variable	<code>flow_lpm</code> . It stores the current recorded pump flow after the latest approved assignment.	If students choose <code>planned_flow_lpm</code> , ask whether the value has been approved/recorded yet.	recognize
1	Planned-flow variable	<code>planned_flow_lpm</code> . It stores a proposed flow before approval.	If students choose <code>gap_lpm</code> , separate flow value from difference-to-target value.	recognize
1	After Cell A state	<code>flow_lpm</code> = 18.5; <code>gap_lpm</code> = 1.5.	If student writes 20.0, distinguish target from current flow.	trace; predict
1	After Cell B current/planned distinction	Current recorded flow remains 18.5; planned flow is 19.5. Planned gap is 0.5 if included.	Key misconception: planned value automatically becomes current. Ask where <code>flow_lpm</code> is reassigned.	trace; explain
1	Cell C approval assignment	<code>flow_lpm</code> = <code>planned_flow_lpm</code> . This makes the first planned value current.	If student names <code>gap_lpm</code> = <code>planned_gap_lpm</code> , note that this updates the gap but not the current flow.	trace; implement
1	After Cell D state	<code>flow_lpm</code> = 19.5; <code>planned_flow_lpm</code> = 20.0; <code>gap_lpm</code> = 0.0. The zero gap came from planned flow, not current recorded flow.	Hardest point: <code>gap_lpm</code> can be based on planned flow while <code>flow_lpm</code> is still old.	trace; explain
1	Debug lab-log claim	Unsafe/incomplete: after Cell D, 20.0 is planned, not current. Current <code>flow_lpm</code> is still 19.5; <code>gap_lpm</code> = 0.0 was calculated from <code>planned_flow_lpm</code> . The checkpoint should say Cell D planned value or wait until Cell E.	Require current/planned/checkpoint language, not just “wrong.”	debug; explain
1	Non-mutating check	Accept <code>print(flow_lpm)</code> or bare <code>flow_lpm</code> . Do not accept an assignment.	If student writes <code>flow_lpm</code> = ..., mark as mutating, not checking.	test; debug
1	Approval assignment completion	<code>planned_flow_lpm</code> . Full line: <code>flow_lpm</code> = <code>planned_flow_lpm</code> .	If student writes a number only, partial credit only if local grading allows; variable name is the transfer target.	implement; trace
1	Transfer/checkpoint explanation	Secure response: checkpoint names prevent copying a planned or stale value as if it were current; the log should say whether the value came after Cell D planning or after Cell E approval/recording.	If generic, ask student to name which cell made 20.0 current.	transfer; explain

4 Expected trace table

Checkpoint	flow_lpm	planned_flow_lpm	gap_lpm	Reason
After Cell A	18.5	not defined	1.5	Baseline gap is 20.0 - 18.5.
After Cell B	18.5	19.5	1.5	First planned correction calculated; current flow not yet reassigned.
After Cell C	19.5	19.5	0.5	Planned flow and planned gap are approved into current variables.
After Cell D	19.5	20.0	0.0	Fine adjustment is planned and the gap is calculated from planned flow, but current flow is still 19.5.
After Cell E	20.0	20.0	0.0	Approved planned flow becomes current.

5 Mastery interpretation and support routing

Band	Point guide	Evidence pattern	Next support action
Secure	8–10 / 10, with trace/debug evidence present	Student can distinguish current versus planned state, identify the approval assignment, and explain why checkpoint language matters.	Continue to Day 3 type/string/casting with one current/planned warm-up.
Developing	5–7 / 10, or uneven tag evidence	Student calculates many values but may treat planned flow or planned gap as current without naming the checkpoint.	Target trace/debug tags; rerun a two-cell planned/current mini-trace.
Needs support	0–4 / 10, or no usable trace/debug evidence	Student cannot yet identify which variable changes on each assignment line.	Model a smaller assignment/overwrite example and complete first trace row together.

6 Optional recovery mini-surface

Use if several students treat a planned value as current before the assignment runs.

```
current = 8
planned = current + 2
# not approved yet
current
current = planned
current
```

Secure recovery answer: before approval, `current` is still 8 even though `planned` is 10. Only `current = planned` makes 10 the current value.

7 Release/accessibility notes

Local verification should check tagged PDF output, extractable text, page count, print-safe margins, render sheets, answer-key separation, no private/protected markers, and the student-facing no-internal-surface rule. If selected for Canvas use, Lance uploads/stages the student PDF and retrieves both Ally and Panorama scores. Working release threshold remains Ally at least 95% and Panorama at least 95%.